$$[\mathrm{Fib}\kappa, K(B,3)]_* = 0$$

Purpose. Informal proof for §3b of `doc/pi3-s2-postnikov-handoff-6.txt`: for $P = \mathrm{Fib}\kappa$ and every abelian group B, the set $[P, K(B,3)]_*$ is a point. Written before the formal proof; numbered statements are meant to be declarations. Foundations as in the development (univalence, images; function extensionality derived); no new axiom; the sphere axiom is not used. *Refereed* independently twice (a fresh reader given the note and the library, not the conclusion; and a second reader): correct, with fixes now incorporated (the name `STr.Q.relates`; the orientation of Lemma 3(a) and of the homotopy in Proposition 6, a `sym` each; the endpoint generalisation in Lemma 3(b); the truncation index in Lemma 8, passed explicitly; the pointing ζ in Proposition 7 given directly; a shared spreading lemma for Steps 1–3 of Propositions 6 and 7). *Status: checked* (`sag/pi3-s2-postnikov/FibreH3.seg`, §8).

Hypotheses. $\kappa : K(\mathbb{Z},2) \to_* K(\mathbb{Z},4)$ is any pointed map for which $\Phi_B : \mathrm{Hom}(\mathbb{Z},B) \to [K(\mathbb{Z},2), K(B,4)]_*$, $g \mapsto [\kappa; K(g,4)]$ is injective for every B (we use only the injectivity in (H4), `KZ2Cohomology.H4`); it is instantiated with `CupSquare.kappa` and `H4Proof.h4`. (H3) (`H3Proof.h3`) is used for the base.

The idea. Classically: the Serre sequence $0 = H^3(K(\mathbb{Z},2)) \to H^3(P) \to H^3(K(\mathbb{Z},3)) \xrightarrow{\tau} H^4(K(\mathbb{Z},2))$, $\tau = \Phi_B$. We prove the two facts that are needed, for this fibration only, with explicit null-homotopies: (i) if the restriction of $F : P \to_* K(B,3)$ to the fibre over the base point is $K(g,3)$, then $\kappa; K(g,4)$ is null, by a homotopy built from F (Proposition 6: this is "τ(restriction of F) $= 0$", and it identifies the connecting map with Φ_B *by construction*: no separate connecting map is defined); (ii) if the restriction is null, F factors through $\mathrm{pr} : P \to K(\mathbb{Z},2)$ (Proposition 7). Exactness of the sequence at $H^3(K(\mathbb{Z},3))$ is not needed.

1. Conventions

As in `doc/h4.pdf`: `isTrunc n X` is "X is $(n-2)$-truncated", `isConn n X` is "X is $(n-2)$-connected"; $K_m = K(\mathbb{Z},m)$, $K_m^B = K(B,m)$ (`EMMaps.KK`), base points $*$; $p \cdot q$ is `trans p q` (by J on q, so $p \cdot \mathsf{refl} \equiv p$); $\Omega Y = (y_0 = y_0, \mathsf{refl})$; $F; G$ is `ptcomp F G` (F first); 0 is `ptconst`; ΩF is `Maps.loop F` with $(\Omega F)(q) = F_0^{-1} \cdot (\mathsf{ap}_F q \cdot F_0)$ (`loop-ap`). $e_m : \Omega K_{m+1}^B \simeq_* K_m^B$ (`KK.e`).

Fix B, and write $W = K_4^B$, $w_0 = *$. Then

$$P = \mathrm{Fib}\kappa = (x : K_2) \times T_x, \quad T_x = (\kappa x = *), \qquad p_0 = (*, \kappa_0),$$

with $\kappa_0 : \kappa * = *$ the pointing of κ. Levels: K_2 is $\mathsf{isConn}\,3$; K_4 is $\mathsf{isConn}\,5$, so each T_x is $\mathsf{isConn}\,4$ (`conn-id`); W is $\mathsf{isTrunc}\,6$, so ΩW and each $(G(\kappa x) = w_0)$ are $\mathsf{isTrunc}\,5$; W is $\mathsf{isConn}\,5$, so ΩW is $\mathsf{isConn}\,4$ and $\Omega^2 W$ is $\mathsf{isConn}\,3$ (both connected).

We work with targets ΩW instead of K_3^B; the two are pointed equivalent by e_3, and statements about classes transfer along a pointed equivalence by pointed-equivalence induction (`ptEqInd`).

2. Three small lemmas

Lemma 1 (constancy is a property)**.** *Let T be $\mathsf{isConn}\,4$, Z be $\mathsf{isTrunc}\,5$ and $d : T \to Z$. Then*

$$\mathsf{Const}(d) = (z : Z) \times (\lambda_\,.\, z = d)$$

is a proposition.

Date: 28 October 2026.

Proof. $\mathsf{Const}(d)$ is the fibre over d of $\mathrm{c} : Z \to (T \to Z)$, $z \mapsto \lambda_.z$; so it suffices that c is an embedding (`Groupoid.emb-fib-prop`). For $z, z' : Z$, $\mathsf{happly} \circ \mathsf{ap}_{\mathrm{c}}$ is, by J, the constant map $(z = z') \to (T \to (z = z'))$, an equivalence by `ConnectedMaps.null` (T is $\mathsf{isConn}\,4$, $z = z'$ is $\mathsf{isTrunc}\,4$); happly is an equivalence; so ap_{c} is one (`Whitehead.cancel-equiv`). □

Here the index matters: T is 2-connected and Z is 3-truncated, so maps $T \to Z$ need not be constant; only the *defect* of a map lies in the 2-truncated $z = z'$, and that is what is used.

Lemma 2 (pointing irrelevance, mere)**.** *If $\Omega(Z, z_0)$ is connected, $k, k' : X \to Z$, $e : k = k'$, $\zeta : k\,x_0 = z_0$, $\zeta' : k' x_0 = z_0$, then $\|(k, \zeta) = (k', \zeta')\|$.*

Proof. J on e; then $\zeta' = \zeta \cdot (\zeta^{-1} \cdot \zeta')$ and $\zeta^{-1} \cdot \zeta'$ is merely refl (as `H4Classes.pi-set`, without the set hypothesis, since the goal is a mere proposition). □

Lemma 3 (two J computations)**.** *Let Y be a type, $y, y_0 : Y$, $k : y = y_0$.*

(a) *If $h : (y = y_0) \to C$, $c : C$ and $h(k \cdot q) = c$ for all $q : y_0 = y_0$, then $h\,t = c$ for all $t : y = y_0$.*
(b) *Let $G : (Y, y_0) \to_* (W', w')$ and $h : (y = y_0) \to (w' = w')$ with $h(k \cdot q) = (\Omega G)(q)$ for all q. Then for all $t : y = y_0$,*

$$\mathsf{ap}_G k \cdot G_0 \;=\; (\mathsf{ap}_G t \cdot G_0) \cdot (h\,t)^{-1}.$$

Proof. J on k (y free; h and the hypothesis in the motive). At $k = \mathsf{refl}$: (a) $h\,t = h(\mathsf{refl} \cdot t) = c$ (`trans-refl-l`). (b) With $u = \mathsf{ap}_G t \cdot G_0$: $h\,t = h(\mathsf{refl} \cdot t) = (\Omega G)(t) = G_0^{-1} \cdot u$. The groupoid identity used is: for $u, v : c = d$,

$$u \cdot (v^{-1} \cdot u)^{-1} = v,$$

by J on u with the endpoint d free and v generalised in the motive (at $d \equiv c$, $u = \mathsf{refl}$: $\mathsf{refl} \cdot (v^{-1} \cdot \mathsf{refl})^{-1} \equiv \mathsf{refl} \cdot (v^{-1})^{-1} = (v^{-1})^{-1} = v$, `trans-refl-l`, `sym-sym`). With $v = G_0$ the right side is $(\mathsf{ap}_G t \cdot G_0) \cdot (h\,t)^{-1}$, and the left side of the claim is $\mathsf{ap}_G \mathsf{refl} \cdot G_0 \equiv \mathsf{refl} \cdot G_0 = G_0$ (`trans-refl-l`). □

3. The fibre restriction

Definition 4. For $F : P \to_* \Omega W$, its restriction to the fibre over $*$ is

$$\rho(F) = (q \mapsto F(*, \kappa_0 \cdot q),\ F_0) \;:\; \Omega K_4 \to_* \Omega W,$$

an element of $\mathtt{PMO} = \mathrm{Map}_*(\Omega K_4, \Omega W)$. (The pointing is F_0 by definition: $\kappa_0 \cdot \mathsf{refl} \equiv \kappa_0$.)

Lemma 5 (the delooping of the restriction)**.** *For each F there is a unique $G_F : K_4 \to_* W$ with $\rho(F) = \Omega G_F$.*

Proof. `DeloopMaps` Corollary 11: $\mathtt{Unloop}\,\rho(F)$ is contractible, with $c = 2$ (K_4 is $\mathsf{isConn}\,4$, by `conn-down`) and W $\mathsf{isTrunc}\,6 = \mathsf{isTrunc}(3+3)$. G_F and the path are the components of its centre. □

4. The transgression of the restriction is zero

Proposition 6 (transgression)**.** *Let $F : P \to_* \Omega W$ and $G : K_4 \to_* W$ with $\rho(F) = \Omega G$. Then $\|0 = \kappa; G\|$ in $\mathrm{Map}_*(K_2, W)$; hence $[\kappa; G] = [0]$.*

Proof. For $x : K_2$ let $Z_x = (G(\kappa x) = w_0)$ and

$$d_x : T_x \to Z_x, \qquad d_x(t) = (\mathsf{ap}_G t \cdot G_0) \cdot F(x, t)^{-1}.$$

Step 1. $\mathsf{Const}(d_x)$ is a proposition (Lemma 1: T_x is $\mathsf{isConn}\,4$, Z_x is $\mathsf{isTrunc}\,5$). *Step 2.* $\mathsf{Const}(d_*)$ holds, with $z = \mathsf{ap}_G \kappa_0 \cdot G_0$: apply Lemma 3(b) with $y = \kappa *$, $k = \kappa_0$, $h = F(*, -)$; its hypothesis $F(*, \kappa_0 \cdot q) = (\Omega G)(q)$ is happly of $\rho(F) = \Omega G$ (first components; by Definition 4, $\rho(F)(q)$ is $F(*, \kappa_0 \cdot q)$); then function extensionality. *Step 3.* By `conn-elim` (K_2 is 0-connected, Step 1, Step 2), $\mathsf{Const}(d_x)$ for every x. Let $N_x : G(\kappa x) = w_0$ be its first component. *Step 4.* N is a

homotopy from the underlying map $x \mapsto G(\kappa x)$ of $\kappa; G$ to $\lambda_.w_0$; by Lemma 2 (ΩW is connected) $\|0 = \kappa; G\|$, and `STr.relates` gives $[0] = [\kappa; G]$. □

Note that only the first component N_x is used: the pointing of the null-homotopy is irrelevant by Lemma 2. What is used about F is exactly that d_x is constant on each fibre: the constancy is a proposition, holds over $*$ by the hypothesis on $\rho(F)$, and spreads along the connected base. This is the transgression: d_x compares the fibre map $F(x,-)$ with the canonical map $t \mapsto \mathsf{ap}_G t \cdot G_0$ from T_x to the fibre Z_x of W, and its constant value trivialises Z_x.

5. Classes with null restriction come from the base

Proposition 7 (exactness at $H^3(P)$)**.** *If $F : P \to_* \Omega W$ and $\rho(F) = 0$, then $\|0 = F\|$.*

Proof. Step 1. For each x, $\mathsf{Const}(F(x,-))$, with $F(x,-) : T_x \to \Omega W$, is a proposition (Lemma 1; ΩW is $\mathsf{isTrunc}\,5$). *Step 2.* $\mathsf{Const}(F(*,-))$ holds with $z = \mathsf{refl}$: Lemma 3(a) with $k = \kappa_0$, $c = \mathsf{refl}$, the hypothesis being happly of $\rho(F) = 0$. *Step 3.* By `conn-elim`, $\mathsf{Const}(F(x,-))$ for all x; let $H\,x : \Omega W$ be its first component, so $H\,x = F(x,t)$ for all t. *Step 4.* The pointing is given directly: $\zeta = (H * = F(*,\kappa_0)) \cdot F_0 : H * = \mathsf{refl}$ (the constancy witness at $t = \kappa_0$, then the pointing of F, since $p_0 = (*,\kappa_0)$); $(H,\zeta) : K_2 \to_* \Omega W$. By (H3), transferred along $e_3^{-1} : K_3^B \simeq_* \Omega W$, $[K_2, \Omega W]_*$ is a point, so merely $e : 0 = (H,\zeta)$ (`effective`). *Step 5.* From e, $\mathsf{refl} = H\,x$ for all x, so $\mathsf{refl} = F(x,t)$ for all (x,t), i.e. $\lambda_.\mathsf{refl} = F_1$; by Lemma 2 ($\Omega^2 W$ is connected), $\|0 = F\|$. □

6. Coefficients: every $K_4 \to_* W$ is some $K(g,4)$

Lemma 8 (climbing)**.** *For $m \geq 1$ and $G : K_{m+1} \to_* K^B_{m+1}$ let $G^\flat = e_m^{-1}; \Omega G; e_m : K_m \to_* K^B_m$ (`unconj`). Then $\mathsf{conj}_m(G^\flat) = \Omega G$, where $\mathsf{conj}_m(F) = e_m; F; e_m^{-1}$ (`Lev.conj`). If $G^\flat = K(g,m)$, then $K(g,m+1) = G$.*

Proof. The first equation: associativity, `pt-inv-l` twice, the unit laws (as `H4Bilinear.H2.conj-unconj`, which is the case $m = 1$). Then $\Omega K(g,m+1) = \mathtt{conj}_m(K(g,m))$ (`Lev.loopK`) $= \mathtt{conj}_m(G^\flat) = \Omega G$, and `unique-loop` at some c with K_{m+1} $\mathsf{isConn}(c+2)$ and K^B_{m+1} $\mathsf{isTrunc}(2c+2)$: for $m = 2$, $c = 2$ and $\mathsf{isTrunc}\,5 \Rightarrow \mathsf{isTrunc}\,6$ (`trunc-suc`); for $m = 3$, $c = 2$ with `conn-down` ($\mathsf{isConn}\,5 \Rightarrow \mathsf{isConn}\,4$) and $\mathsf{isTrunc}\,6$ exactly. The lift is a step: $\mathsf{isTrunc}(m+3)$ is not literally $\mathsf{isTrunc}(2c+2)$. □

Corollary 9. *Every $G : K_4 \to_* K_4^B$ is $K(g,4)$ for $g = \mathrm{inv}_4(G)$, where $\mathrm{inv}_4(G) = \mathrm{inv}_2(((G^\flat)^\flat))$ and inv_2 is `H4Bilinear.H2.inv2` (`k2-surj`: $K(\mathrm{inv}_2 F, 2) = F$).*

Proof. Lemma 8 at $m = 2$, then at $m = 3$. □

7. The theorem

Theorem 10. *For every abelian B, $[P, K_3^B]_*$ is contractible, with centre $[0]$.*

Proof. By transport along $e_3 : \Omega W \simeq_* K_3^B$ it suffices to treat $[P, \Omega W]_*$. Classes are eliminated into the propositions $[0] = c$ (`STr.ind`); for $F : P \to_* \Omega W$ we show $\|0 = F\|$. Let $G = G_F$ (Lemma 5). By Proposition 6, $[\kappa; G] = [0]$. By Corollary 9, $G = K(g,4)$ and $0 = K(g',4)$ with $g = \mathrm{inv}_4 G$, $g' = \mathrm{inv}_4 0$. Then

$$\Phi_B(g) = [\kappa; K(g,4)] = [\kappa; G] = [0] = [\kappa; 0] = [\kappa; K(g',4)] = \Phi_B(g')$$

(`Lev.K-def` for the first and last steps, as in `H4Proof.phi-diag`; `comp-const-r` for $[0] = [\kappa; 0]$). By injectivity of Φ_B, $g = g'$, so $G = K(g,4) = K(g',4) = 0$ and $\rho(F) = \Omega G = \Omega 0 = 0$ (`loop-const`). Proposition 7 gives $\|0 = F\|$. □

Remark 11. (1) The connecting map is never defined separately: the step "$\tau(\rho F) = \Phi_B(g)$" is Proposition 6 applied to $G = K(g,4)$, and the class it produces is literally $[\kappa; K(g,4)]$, the value of `KZ2Cohomology.Phi`. (2) Only injectivity of Φ_B is used, and only (H3) for $K(\mathbb{Z},2)$; no statement about $H^3(K(\mathbb{Z},3);B)$ beyond Lemma 5 and Corollary 9. (3) Nothing is said about the type of null-homotopies; all conclusions are mere. (4) The same proof works for any κ with Φ_B injective; the module is parameterised by κ and a proof of (H4).

8. Correspondence with the formal proof

Status: checked (69 modules, 0 failed; a fresh process: 2877 declarations, 1.1 s; every `FibreH3` declaration under 0.01 s; `fibre.opens`, 84 entries, every line "nothing else opened"). Module `FibreH3`: `Const`, `const-emb`, `const-prop` (Lemma 1); `pi-mere` (Lemma 2); `reparam`, `cancel-v`, `TG.tg` (Lemma 3); `Climb[A,B]`: `unconj`, `conj-unconj`, `climb`, `inv3`, `surj3`, `inv4`, `surj4` (Lemma 8, Corollary 9); `spread` (Steps 1–3 of Propositions 6, 7); `FibAt[kappa, h4k, B]`: `T`, `cT`, `cOW`, `cOOW` (levels), `res`, `res-def` (Definition 4), `deloop-c`, `deloop`, `deloop-eq` (Lemma 5), `Trans[F,G,w]`: `Z`, `d`, `hyp`, `const-base`, `Nu` (N_x), `null`, `trans-cls` (Proposition 6); `h3-loop` ((H3) transferred); `Exact[F,z]`: `hyp`, `const-base`, `cF`, `H`, `H-eq`, `zeta`, `base-zero`, `from-base`, `exact` (Proposition 7); `g0`, `Kill[F]`: `g`, `cls-eq`, `g-eq`, `G-zero`, `res-zero`, `merely-zero`, `contr-loop`, `contr` (Theorem 10); `fib-h3 : (B : Ab) -> isContr [Fib kappa, K(B,3)]` for `CupSquare.kappa` and `H4Proof.h4`.